2026年中国地球科学联合学术年会

2026-10-20 · 专题报告安排

第149专题 Advances in Space Weather/Environment Hazards Powered by Supercomputing, Intelligent Computing, and AI Technologies | 口头报告

2026-10-20 第39会议室,Room 39(Fourth Floor,403)

主持人:Mei Li, Hailong Yang, Gang Qin, Xianger Lei

主持人:Zhengkun Qin, Liang Dong, Dingwen Tao, Hualing Wei

时间	序	报告题目	报告人
08:30-08:55	*1	Spatio-temporal features of seismo-ionospheric influence and its potential LAI coupling mechanism	Mei Li(李美 CHAIR)
08:55-09:10	2	Post-Earthquake Real-Time Seismic Simulation on the Line Shine Exascale Supercomputer: Methodology and Applications	Wenqiang Wang
09:10-09:25	3	Adapting HPC Software to Hardware with Effective Performance Analysis and Optimization	Hailong Yang((杨海龙) CHAIR)
09:25-09:40	4	GPU Parallel Implementation of Lattice Boltzmann Method for Magnetohydrodynamic Flow	Wu Wang(王武)
09:40-09:55	5	Integrating the Intelligent Computing Ecosystem to Empower Earth Sciences: An Innovative Exploration of China’ s National Supercomputing Internet(融通智算生态，算赋地球科学—国家超算互联网的创新探索)	Jingyu Liu(刘靖宇)
09:55-10:10	6	Counterclockwise rotation of sunspot AR 4482 and its physical implications	Delong Chi
10:10-10:20		BREAK	
10:20-10:45	*7	Solar Energetic Particles and Cosmic Rays: Effects on the Space Environment(太阳高能粒子和宇宙线对空间环境影响研究)	Gang Qin(秦刚 CHAIR)
10:45-11:00	8	Kinetic Modeling of Coherent Radio Emission from Flare-Energetic Electrons in Coronal Loops	Mehdi Yousefzadeh
11:00-11:15	9	Validating the Mg I 12.32 μm line Diagnostics for Upper-Photospheric Magnetism	Mohamed Sedik
11:15-11:30	10	A Multiphysics Computational Framework Integrating Computational Fluid Dynamics and Electromagnetic Modeling for Complex Transport Phenomena	Walaa Elhamamy(李惟清)
11:30-11:45	11	The Effect of Plasma-β on the Heating Mechanisms during Magnetic Reconnection in Partially Ionized Low Solar Atmosphere	Abdullah Zafar
11:45-12:00	12	Discussion on the Variation of the Cosmic Gamma Ray Flux Observed by LHAASO during Solar Activity	Xianger Lei(CHAIR)
时间	序	报告题目	报告人
13:30-13:55	*13	Physics-Driven Deep Learning Reconstruction of FY-3D MWRI Brightness Temperatures Incorporating Static and Dynamic Constraints(融合静动态物理约束的 FY-3D MWRI 亮温物理驱动深度学习重构算法研究)	Zhengkun Qin(秦正坤 CHAIR)
13:55-14:10	14	Magnetosphere-Dynamo Coupling at Mercury: Insights from Bow Shock and Magnetopause Responses to Solar Wind Forcing	Peishan He
14:10-14:25	◎15	Research Progress on SEPs on the Fine Structures of the Large Temporal-spataial Current Sheets in Solar Flares/CMEs	Zhihe Wang(王智禾)

时间	序	报告题目	报告人
14:25-14:40	16	Lithosphere-Atmosphere-Ionosphere Coupling: Opportunities and Challenges from Phenomenon Observation to Physical Prediction	Wang Li(李旺)
14:40-14:55	17	Progress in China–Malaysia Collaborative Research on Space Weather and Radio Astronomy(中马联合空间天气/射电天文研究进展介绍)	Liang Dong(CHAIR)
14:55-15:10	18	Patents and Software Copyrights: Application, Examination, Commercialization, and Future Prospects(专利与软著申请、受理、转化介绍及未来展望)	Huanxin Feng(冯焕新)
15:10-15:20		BREAK	
15:20-15:45	*19	Error-Bounded Lossy Compression and Parallel I/O Acceleration for High-Fidelity Space Weather Simulations (面向空间天气高保真数值模拟的误差可控有损压缩与并行 I/O 加速)	Dingwen Tao(陶鼎文 CHAIR)
15:45-16:00	20	Causal Analysis and Predictive Modelling of Space Weather Using Information-Theoretic Measure, Nonlinear System Identification and Machine Learning Techniques	Hualing Wei(CHAIR)
16:00-16:15	21	A Hierarchical Hydrological–Hydraulic–CFD Framework for Urban Flood Inundation Assessment Under Typhoon-Induced Extreme Rainfall	Walaa Elhamamy(李惟清)
16:15-16:30	22	Isotopic Fractionation in Solar Energetic Particle Acceleration: Evidence from Resonant Wave–Particle Interactions and Turbulent Processes	Bojing Zhu(朱伯靖)
16:30-16:45	◎23	The Mystery of Helium-3 Enrichment in Solar Energetic Particle Acceleration	Zehua Fu(付泽华)
16:45-17:00	24	Long-Term Influence of Solar High-Energy Particles and Cosmic Rays on Lunar Surface Evolution: Modeling with the RHPIC–LBM Framework	Henry Tufo

来源：https://www.cgu.org.cn/cugs/?q=node/109&subject=149&list=12

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逐人起止时间未公布；以下保留原文顺序。参考时段：原文未公布

序	报告题目	报告人
*330	Electromagnetic Wave Propagation Characteristics Across the Coupled Earth–Space System Using the Lattice Boltzmann Method	Pengxiang Yang, Yongbing Li
◎331	Numerical detection methods for Type II and Type III radio bursts using radio telescope observations	Zhen Kou
*◎332	Progress in Identifying Solar Energetic Particle Sources and Origins: Implications for Modeling and Predicting Space Weather Disasters	Zhihe Wang,Henry Tufo
333	Triggering of Earthquakes by Strong Fluctuations of the Interplanetary Magnetic Field	Delong Chi
334	Primary Causes of Venus’ s Anomalously High Surface Temperature and Global Warming	Delong Chi
*◎335	An improved RHPIC-LBM input module for SEP space weather modeling and forecasting driven by large-scale, multi-layered, ground-to-space 3D observations	Zhenkou, Henry Tufo

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